

PHL1010: Practice Problems III
(03/11/2025)

- A long circular cylinder of radius R carries a magnetization $\mathbf{M} = kr^2\hat{\phi}$ where k is a constant, r is the distance from the axis. Find the magnetic field due to $\mathbf{M}$, for points inside and outside the cylinder.
- Suppose the field inside a large piece of magnetic material is $\mathbf{B}_0$, so that $\mathbf{H}_0 = (1/\mu_0)\mathbf{B}_0 - \mathbf{M}$, where $\mathbf{M}$ is a “frozen-in” magnetization.
 - Now a small spherical cavity is hollowed out of the material (Fig. 1). Find the field at the center of the cavity, in terms of $\mathbf{B}_0$ and $\mathbf{M}$. Also find $\mathbf{H}$ at the center of the cavity, in terms of $\mathbf{H}_0$ and $\mathbf{M}$.
 - Do the same for a long needle-shaped cavity running parallel to $\mathbf{M}$.
 - Do the same for a thin wafer-shaped cavity perpendicular to $\mathbf{M}$.
- A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating material of magnetic susceptibility χ_m . A current I flows down the inner conductor and returns along the outer ones; in each case the current distributes itself uniformly over the surface (Fig. 2). Find the magnetic field in the region between the tubes. As a check. Calculate the magnetization and the bound currents and confirm that they (together with free current) generate the correct field.
- Notice the following parallel:

$$\nabla \cdot \mathbf{D} = 0, \nabla \times \mathbf{E} = 0, \epsilon_0 \mathbf{E} = \mathbf{D} - \mathbf{P} \quad (\text{no free charge});$$

$$\nabla \cdot \mathbf{B} = 0, \nabla \times \mathbf{H} = 0, \mu_0 \mathbf{H} = \mathbf{B} - \mu_0 \mathbf{M} \quad (\text{no free current}).$$
 Thus, the transcription $\mathbf{D} \rightarrow \mathbf{B}, \mathbf{E} \rightarrow \mathbf{H}, \mathbf{P} \rightarrow \mu_0 \mathbf{M}, \epsilon_0 \rightarrow \mu_0$ turns an electrostatic problem into an analogous magnetostatic one. Use this, together with your knowledge of the electrostatic results, to re-derive
 - the magnetic field inside a uniformly magnetized sphere,
 - the magnetic field inside a sphere of linear magnetic material in an otherwise uniform magnetic field.
- At the interface between one linear magnetic material and another, the magnetic field lines bend (Fig. 3). Show that $\tan \theta_2 / \tan \theta_1 = \mu_2 / \mu_1$, assuming there is no free current at the boundary. What do you expect if μ_2 is very large and μ_1 is very small?

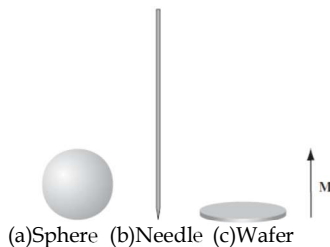


Figure 1

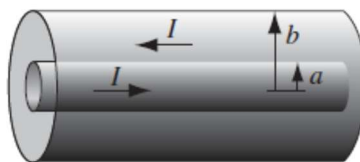


Figure 2

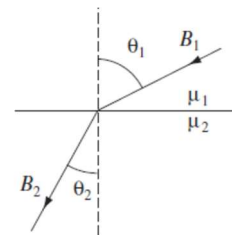


Figure 3

6. A square loop of wire (side a) lies on a table, a distance s from a very long straight wire, which carries a current I , as shown in Fig. 4.
- Find the flux of $\mathbf{B}$ through the loop.
 - If someone now pulls the loop directly away from the wire, at speed v , what emf is generated? In what direction (clockwise or counter-clockwise) does the current flow?
 - What if the loop is pulled to the *right* at speed v ?
7. A square loop is cut out of a thick sheet of aluminum. It is then placed so that the top portion is in a uniform magnetic field $\mathbf{B}$, and is allowed to fall under gravity (Fig. 5). (In the diagram, shading indicates the field region; $\mathbf{B}$ points into the page.)
- Find the terminal velocity of the loop.
 - Find the velocity of the loop as a function of time.
 - How long does it take (in seconds) to reach, say, 90% of the terminal velocity?
8. A perfectly conducting spherical shell of radius a rotates about the z axis with angular velocity ω , in a uniform magnetic field $\mathbf{B} = B_0 \hat{z}$. Calculate the emf developed between the "north pole" and the equator.
9. Two tiny wire loops, with area $\mathbf{a}_1$ and $\mathbf{a}_2$, are situated a displacement $\mathbf{r}$ apart (Fig. 6). (a) Find their mutual inductance? (b) Suppose a current I_1 is flowing in loop 1, and we propose to turn on current I_2 in loop 2. How much work must be done, against the mutually induced emf, to keep the current I_1 flowing in loop 1?

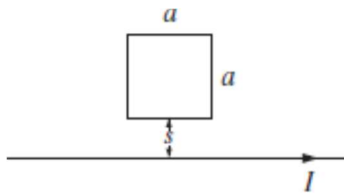


Figure 4

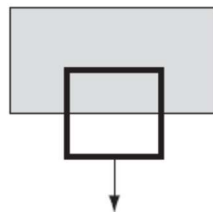


Figure 5

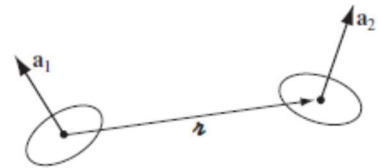


Figure 6